Study Protocol

Effect of 12-Week Time-Restricted Eating on Glycemic Control in Prediabetic Adults: A Synthetic Pilot Study

Protocol ID: SPbE-2026 | Version 1.0 | Synthetic / fictional data for software testing only.

1. Background and Rationale

Prediabetes affects a large fraction of middle-aged adults and is a major risk factor for progression to type 2 diabetes. Time-restricted eating (TRE), in which daily food intake is confined to a fixed window, has been proposed as a low-cost behavioral intervention to improve glycemic control. This synthetic pilot study models a randomized comparison of TRE versus an unrestricted-schedule control over 12 weeks. *All subjects and results in this protocol are fabricated for the purpose of testing AI-assisted metadata generation.*

2. Objectives

Primary objective: estimate the effect of a 10-hour TRE window on change in fasting plasma glucose from baseline to week 12. Secondary objectives: change in HbA1c, body weight, blood pressure, and lipid panel; protocol adherence; and safety.

3. Study Design

- Design: two-arm, parallel-group, randomized pilot.
- Population: adults aged 40-65 with prediabetes (fasting glucose 100-125 mg/dL or HbA1c 5.7-6.4%).
- Arms: TRE (10-hour eating window, 08:00-18:00) vs. control (habitual eating schedule).
- Duration: 12 weeks. Visits at baseline and week 12.
- Target enrollment: 40 subjects (1:1 allocation).

4. Eligibility

Inclusion: age 40-65; BMI 26-35 kg/m^2; documented prediabetes; able to provide informed consent. **Exclusion:** diagnosed diabetes; current use of glucose-lowering medication; pregnancy; active eating disorder; shift work preventing a fixed eating window.

5. Measurements

Fasting plasma glucose, HbA1c, body weight, BMI, systolic and diastolic blood pressure, and a fasting lipid panel (LDL, HDL, triglycerides) were recorded. Adherence was captured as the percentage of days the assigned eating window was followed, from participant diaries. All variables and their units are defined in the accompanying data dictionary.

6. Statistical Analysis

The primary endpoint (change in fasting glucose) will be compared between arms using a two-sample t-test, with a linear model adjusting for baseline value, age, and sex as a sensitivity analysis. Given the pilot sample size, results are intended to inform feasibility and effect-size estimates rather than confirmatory inference.

7. Ethics and Data Sharing

As a synthetic dataset, no human subjects were involved and no ethics approval applies. The data are released to support FAIR-metadata tooling and may be freely reused. Metadata for the study and its datasets are described using CEDAR templates and registered in the Canopy data hub.